Detecting Spills Using Thermal Imaging, Pretrained Deep Learning Models, and a Robotic Platform

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ABSTRACT

Undetected spills in public spaces such as cafes, restaurants, and retail stores are a leading cause of slip-and-fall injuries. This work presents a real-time binary spill detection system that combines pretrained deep convolutional neural networks with thermal (infrared) imaging deployed on a consumer-grade robotic platform.

Using a balanced dataset of 4,000 images (RGB and thermal, spill and no-spill), we demonstrate that thermal-based models outperform RGB-based counterparts in accuracy, inference speed, and model compactness. Our best configuration — a VGG19 classifier trained on thermal images — achieves 100% accuracy on test data and in live deployment, with inference times as low as 44 ms and a model size of 324.6 MB.

METHODOLOGY

Images were captured simultaneously using a Topdon TC001 thermal camera and a Genius WideCam F100 RGB camera mounted on a mobile robot platform, controlled via USB joystick over serial communication. A synchronized Python script handled dual-camera capture. All training and inference was performed on a Lenovo Legion Pro 7i with an NVIDIA RTX 4080.

Data was collected across two rooms — an Atrium with porcelain tile flooring under consistent lighting, and room J234 with polished concrete under dynamic natural lighting — and four liquids: water, Coke, red juice, and yellow juice. This yielded 8 environment combinations (Room × Liquid × Modality). Spill sizes ranged from 2–4 inches in diameter (small) to 12+ inch irregular shapes (large). The final dataset totaled 4,000 images, evenly split between modalities (RGB and thermal) and classes (spill and no-spill).

A third modality — side-by-side combined images — was created by horizontally concatenating aligned thermal and RGB frames (thermal left, RGB right) at a resolution of 512×192.
Data was divided into 70% training, 20% validation, and 10% test.
RGB and thermal frames were spatially aligned through cropping and perspective transformation.
All models were fine-tuned using the last 5 layers of pretrained ImageNet weights. Training used an RMSprop optimizer (lr = 1×10^{-5}), binary cross-entropy loss, and early stopping with patience = 5 over a maximum of 50 epochs (never reached). Batch sizes were 8 for training/validation and 2 for testing. Augmentation included random horizontal flipping, slight rotation (factor 0.01), and minor contrast variation (factor 0.01), followed by the standard VGG preprocessing pipeline.

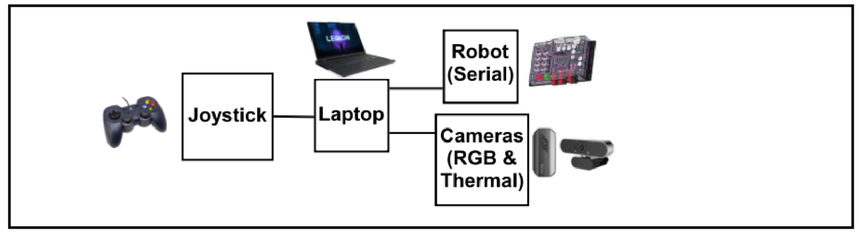
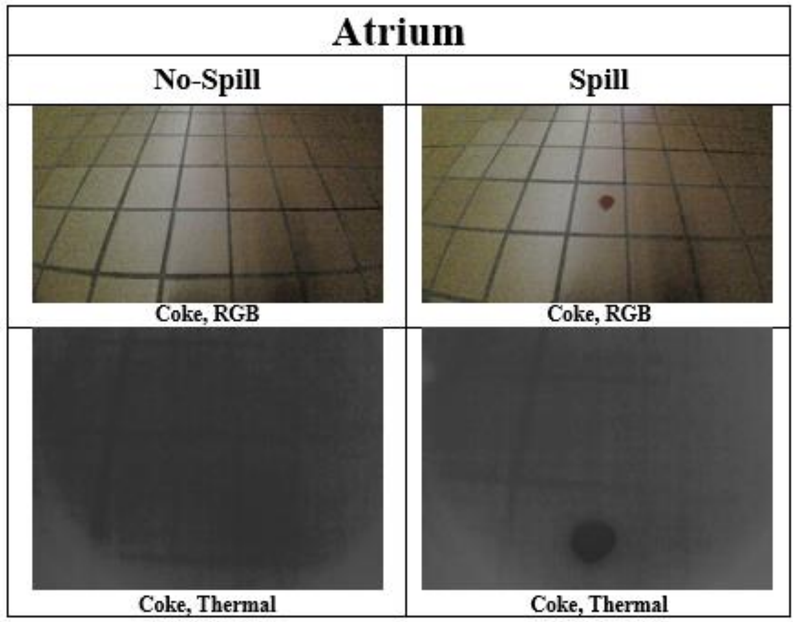


Fig. 1. A diagram of the data collection and testing setup

SAMPLE IMAGES



FUTURE WORK

Future experiments should relax the controlled conditions of this study by introducing environmental heat sources such as foot traffic, nearby heaters, and direct sunlight, variables not accounted for, to isolate spill signatures. Additionally, ensemble or fusion-based methods that jointly process RGB and thermal features are a promising direction: RGB input could correct false positives caused by thermal clutter, while thermal input maintains robustness under poor lighting. Pixel-level segmentation models could extend the system beyond binary classification toward precise hazard localization. Finally, porting the thermal VGG19 model to embedded platforms such as the NVIDIA Jetson would enable fully autonomous robot deployment without tethered compute.

RESULTS AND CONCLUSION

A VGG19 model was trained separately on RGB, thermal, and combined images to evaluate which modality best supports spill detection. Thermal imaging achieved 100% accuracy on both the test dataset and live deployment, with the smallest model size (324.6 MB) and fastest inference time (44 ms). RGB reached 98.84% test accuracy but produced a model three times larger (1.0 GB) with slower inference (55 ms). The combined modality, while 100% accurate on the test set, dropped to only 60% accuracy during real-time demo testing, indicating poor generalization to live conditions.

Individual models trained on each of the 8 room-liquid combinations consistently achieved 100% accuracy on both test sets and real-time deployment, confirming that thermal VGG19 generalizes reliably across floor types, liquids, and lighting conditions — provided environmental heat sources are isolated.

Eleven pretrained architectures were evaluated on the full thermal dataset. Only VGG19 and NasNetMobile achieved 100% test accuracy. VGG19 was selected as the superior choice: it matches NasNetMobile's accuracy while running 9 ms faster on average and requiring 115.7 MB less storage.

In environments with varied lighting and isolation from environmental heat sources, a VGG19 model trained on thermal imaging delivers the best performance across all measured criteria — inference time, test accuracy, and real-time deployment reliability.

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